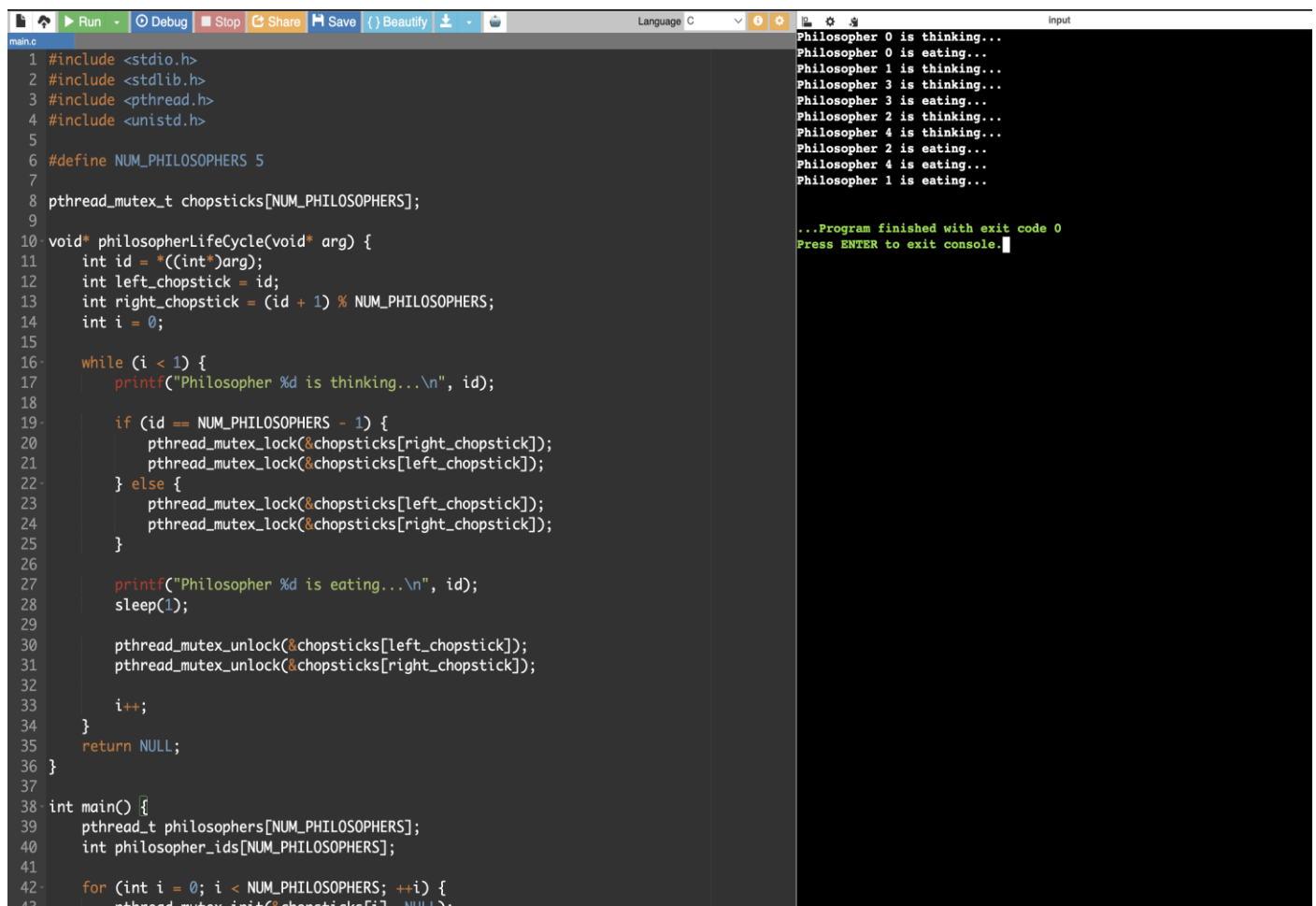


EXPERIMENT 12:

Design a C program to simulate the concept of Dining-Philosophers problem



The image shows a code editor with a C program for the Dining-Philosophers problem. The program defines 5 philosophers and 5 chopsticks. Each philosopher has a lifecycle of thinking, eating, and sleeping. The main function creates an array of philosopher threads and starts them. The output on the right shows the sequence of events: philosophers thinking and eating in a sequence that avoids deadlock.

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <pthread.h>
4 #include <unistd.h>
5
6 #define NUM_PHILOSOPHERS 5
7
8 pthread_mutex_t chopsticks[NUM_PHILOSOPHERS];
9
10 void* philosopherLifeCycle(void* arg) {
11     int id = *((int*)arg);
12     int left_chopstick = id;
13     int right_chopstick = (id + 1) % NUM_PHILOSOPHERS;
14     int i = 0;
15
16     while (i < 1) {
17         printf("Philosopher %d is thinking...\n", id);
18
19         if (id == NUM_PHILOSOPHERS - 1) {
20             pthread_mutex_lock(&chopsticks[right_chopstick]);
21             pthread_mutex_lock(&chopsticks[left_chopstick]);
22         } else {
23             pthread_mutex_lock(&chopsticks[left_chopstick]);
24             pthread_mutex_lock(&chopsticks[right_chopstick]);
25         }
26
27         printf("Philosopher %d is eating...\n", id);
28         sleep(1);
29
30         pthread_mutex_unlock(&chopsticks[left_chopstick]);
31         pthread_mutex_unlock(&chopsticks[right_chopstick]);
32
33         i++;
34     }
35     return NULL;
36 }
37
38 int main() {
39     pthread_t philosophers[NUM_PHILOSOPHERS];
40     int philosopher_ids[NUM_PHILOSOPHERS];
41
42     for (int i = 0; i < NUM_PHILOSOPHERS; ++i) {
43         pthread_mutex_init(&chopsticks[i], NULL);
44         philosopher_ids[i] = i;
45         pthread_create(&philosophers[i], NULL, philosopherLifeCycle, &philosopher_ids[i]);
46     }
47
48     for (int i = 0; i < NUM_PHILOSOPHERS; ++i) {
49         pthread_join(philosophers[i], NULL);
50     }
51
52     return 0;
53 }
```

Philosopher 0 is thinking...
Philosopher 0 is eating...
Philosopher 1 is thinking...
Philosopher 3 is thinking...
Philosopher 3 is eating...
Philosopher 2 is thinking...
Philosopher 4 is thinking...
Philosopher 2 is eating...
Philosopher 4 is eating...
Philosopher 1 is eating...

...Program finished with exit code 0
Press ENTER to exit console.